

DIFFERENTIABILITY

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Lecture Outline

- 1 Introduction
 - Non Differentiable Functions
- 2 Derivatives of elementary functions
- 3 Differentiation Rules
- 4 Implicit Differentiation
- 5 Higher-Order derivatives



Introduction

- 1 The notion of **differentiability** is concerned with describe the rate of change of a quantity relative to another.
- 2 Suppose the relation is expressed as a function then the derivative describes the rate of change of the dependent variable relative to the independent variable.
- 3 We can recall that, the slope/ tangent to the graph of the function at given points is:

$$\text{slope} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} \quad (1)$$

- 4 The slope of the line passing through the point $(x_0, f(x_0))$ and touching the graph of f is

$$\text{Slope} = \frac{f(x) - f(x_0)}{x - x_0}$$



Introduction

Definition

Let $f(x)$ be defined at $x_0 \in (a, b)$. Then the derivative of $f(x)$ at x_0 denoted by $f'(x_0)$ is defined by

$$f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

provided the limit exist.

Suppose x close to x_0 by some small value h , then $x = x_0 + h$ and so we have

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} \quad (2)$$

provided the limit exist.



Introduction

Remarks

- ① $f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$ is the derivative of f at x_0 .
- ② $f'(x_0^-) = \lim_{x \rightarrow x_0^-} \frac{f(x) - f(x_0)}{x - x_0}$ is the **left-derivative** of f at x_0 .
- ③ $f'(x_0^+) = \lim_{x \rightarrow x_0^+} \frac{f(x) - f(x_0)}{x - x_0}$ is the **right-derivative** of f at x_0 .



Introduction

Definition

- 1 **Differentiation** is the process of finding the derivative of a function.
- 2 f is said to be **differentiable** at x_0 when $f'(x_0)$ exists.
- 3 A function $f(x)$ is **differentiable in an interval** (a, b) if it is differentiable at every point in (a, b) .

Notations

The following notations can be used for derivative of $f(x)$

$$1) f'(x), \quad 2) \frac{df(x)}{dx}, \quad 3) D_x f(x), \quad 4) Df(x)$$



Example

Find the derivative of $f(x) = x^2 + 3$ at $x = 2$.

We need to determine $f'(2) = \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h}$. We have

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} \quad (3)$$

$$= \lim_{h \rightarrow 0} \frac{h^2 + 4h + 4 - 4}{h} \quad (4)$$

$$= \lim_{h \rightarrow 0} (h + 4) \quad (5)$$

$$= 4 \quad (6)$$



Example

Show that $f(x) = x^3$ is differentiable at $x = a$.

We need to determine $f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$. We have

$$\begin{aligned} f'(a) &= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(a+h)^3 - a^3}{h} \\ &= \lim_{h \rightarrow 0} \frac{a^3 + 3a^2h + 3ah^2 + h^3 - a^3}{h} \\ &= \lim_{h \rightarrow 0} (3a^2 + 3ah^2 + h^2) \\ &= 3a^2 \end{aligned}$$



Example

Determine the interval within which $f(x) = \sqrt{x-1}$ is differentiable.

We need to determine $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ and examine the domain. We have

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{x+h-1} - \sqrt{x-1}}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\sqrt{x+h-1} - \sqrt{x-1}}{h} \times \frac{\sqrt{x+h-1} + \sqrt{x-1}}{\sqrt{x+h-1} + \sqrt{x-1}} \\
 &= \lim_{h \rightarrow 0} \frac{x+h-1 - (x-1)}{h(\sqrt{x+h-1} + \sqrt{x-1})} \\
 &= \frac{1}{\sqrt{x-1} + \sqrt{x-1}} = \frac{1}{2\sqrt{x-1}}
 \end{aligned}$$



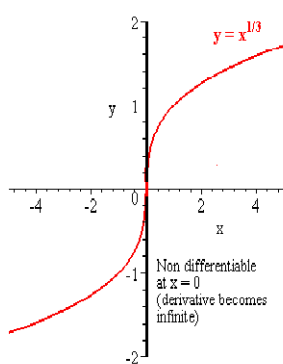
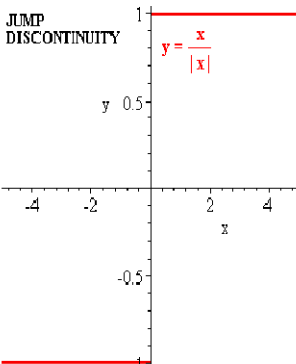
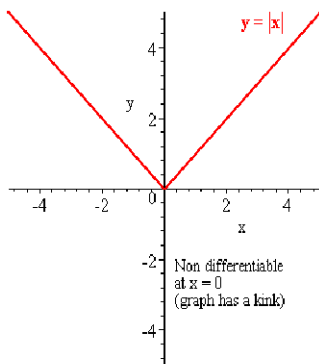
- ① Thus $f'(x)$ is defined if $2\sqrt{x-1} > 0 \implies x > 1$.
- ② Therefore the function $f(x) = \sqrt{x-1}$ is differentiable in $(1, \infty)$.
- ③ This is smaller than the domain of $f(x) = \sqrt{x-1}$ which is $[1, \infty)$.



If f is differentiable at a , then it is continuous at a .

Thus a function is **not differentiable** at a point if

- 1 the graph has corner or a kink
- 2 the function is discontinuous there
- 3 there is a vertical tangent line to the function at that point



Derivatives of elementary functions

We now turn our focus to the derivatives of the following functions:

- ① Constant
- ② Power
- ③ Trigonometric
- ④ Exponential
- ⑤ Logarithmic
- ⑥ Hyperbolic
- ⑦ Inverse Trigonometric
- ⑧ Inverse Hyperbolic



Differentiation of constant functions

The derivatives of constants function $f(x) = c$ are given by

$$\frac{d}{dx}f(x) = 0$$

for c constant. that is

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \frac{c - c}{h} = 0 \quad (7)$$



Differentiation of Power functions

Power functions are functions of the form

$$f(x) = x^n \quad (8)$$

where $n \in \mathbb{R}$. Then we have

$$\frac{d}{dx} f(x) = \frac{d}{dx} x^n \quad (9)$$

$$= nx^{n-1} \quad (10)$$



Example

$$1) f(x) = x^4, \quad 2) f(x) = \frac{1}{x^5}, \quad 3) f(x) = \sqrt[3]{x}$$

$$\textcircled{1} \quad f(x) = x^4 \implies \frac{d}{dx} f(x) = 4x^{4-1} = 4x^3.$$

$$\textcircled{2} \quad f(x) = x^{-5} \implies \frac{d}{dx} f(x) = -5x^{-5-1} = -\frac{5}{x^6}.$$

$$\textcircled{3} \quad f(x) = x^{1/3} \implies \frac{d}{dx} f(x) = \frac{1}{3}x^{(1/3)-1} = \frac{1}{3}x^{-2/3} = \frac{1}{3\sqrt[3]{x^2}}.$$



Differentiation of Trigonometric functions

Below are the derivatives of circular functions

$$① \quad \frac{d}{dx} \sin x = \cos x$$

$$② \quad \frac{d}{dx} \cos x = -\sin x$$

$$③ \quad \frac{d}{dx} \tan x = \sec^2 x$$

$$④ \quad \frac{d}{dx} \cot x = -\csc^2 x$$

$$⑤ \quad \frac{d}{dx} \sec x = \sec x \tan x$$

$$⑥ \quad \frac{d}{dx} \csc x = -\csc x \cot x$$



Differentiation of Inverse Trigonometric Functions

$$① \quad \frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$② \quad \frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$$

$$③ \quad \frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$④ \quad \frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1+x^2}$$

$$⑤ \quad \frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{1-x^2}}$$

$$⑥ \quad \frac{d}{dx}(\csc^{-1} x) = -\frac{1}{|x|\sqrt{1-x^2}}$$



Differentiation of Exponential and Logarithmic functions

- 1 The derivative of functions of the form a^x is given by

$$\frac{d}{dx} a^x = a^x \ln a \quad (11)$$

2

$$\frac{d}{dx} e^x = \frac{d}{dx} (x) \times e^x = e^x \quad (12)$$

3

$$\frac{d}{dx} \ln x = \frac{\frac{d}{dx} (x)}{x} = \frac{1}{x} \quad (13)$$



Differentiation of Hyperbolic Functions

$$① \quad \frac{d}{dx}(\sinh x) = \cosh x$$

$$② \quad \frac{d}{dx}(\cosh x) = \sinh x$$

$$③ \quad \frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$$

$$④ \quad \frac{d}{dx}(\coth x) = -\operatorname{csch}^2 x$$

$$⑤ \quad \frac{d}{dx}(\operatorname{sech} x) = -\operatorname{sech} x \tanh x$$

$$⑥ \quad \frac{d}{dx}(\operatorname{csch} x) = -\operatorname{csch} x \coth x$$



Differentiation of Inverse Hyperbolic Functions

$$① \quad \frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{x^2 + 1}}$$

$$② \quad \frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2 - 1}}; \quad x > 1$$

$$③ \quad \frac{d}{dx}(\tanh^{-1} x) = \frac{1}{1 - x^2}; \quad |x| < 1$$

$$④ \quad \frac{d}{dx}(\coth^{-1} x) = \frac{1}{1 - x^2}; \quad |x| > 1$$

$$⑤ \quad \frac{d}{dx}(\operatorname{sech}^{-1} x) = -\frac{1}{x\sqrt{1 - x^2}}; \quad 0 < x < 1$$

$$⑥ \quad \frac{d}{dx}(\operatorname{csch}^{-1} x) = -\frac{1}{|x|\sqrt{1 - x^2}}, \quad x \neq 0$$



Differentiation of sums of functions

For functions $f(x)$ and $g(x)$, and constants α, β

$$\frac{d}{dx}(\alpha f(x) \pm \beta g(x)) = \alpha \frac{d}{dx} f(x) \pm \beta \frac{d}{dx} g(x) \quad (14)$$

Example

Find the first derivative of $f(x) = x^4 + 6 \sin x + 4 - \ln x$

We have that

$$\begin{aligned} f'(x) &= \frac{d}{dx} x^4 + 6 \frac{d}{dx} \sin x + \frac{d}{dx} 4 - \frac{d}{dx} \ln x \\ &= 4x^3 + 6 \cos x + 0 - \frac{1}{x} \\ &= 4x^3 + 6 \cos x - \frac{1}{x} \end{aligned}$$



Differentiation of products of functions

The derivative of product of functions f and g is given by

$$\frac{d}{dx}fg = f\frac{d}{dx}g + g\frac{d}{dx}f \quad (15)$$

Example

Find $h'(x)$ if $h(x) = \sin x \cos x$

We notice that $h(x)$ is the product of the functions $f(x) = \sin x$ and $g(x) = \cos x$. Hence applying the product rule we have

$$\begin{aligned} \frac{d}{dx}h(x) &= f\frac{d}{dx}g + g\frac{d}{dx}f \\ &= \sin x(-\sin x) + \cos x(\cos x) \\ &= -\sin^2 x + \cos^2 x \end{aligned}$$



Example

Find $h'(x)$ if $h(x) = e^x \csc x$

$$\begin{aligned}
 \frac{d}{dx}h(x) &= f \frac{d}{dx}g + g \frac{d}{dx}f \\
 &= e^x(-\csc x \cot x) + \csc x(e^x) \\
 &= -e^x \csc x \cot x + e^x \csc x \\
 &= e^x \csc x(1 - \cot x)
 \end{aligned}$$

Example

Find $h'(x)$ if $h(x) = (e^x + x^2)(3^x + \tan x)$

$$\begin{aligned}
 \frac{d}{dx}h(x) &= f \frac{d}{dx}g + g \frac{d}{dx}f \\
 &= (e^x + x^2)(3^x \ln 3 + \sec^2 x) + (3^x + \tan x)(2x + e^x)
 \end{aligned}$$



Differentiation of Rational functions

The derivative of rational functions is given by

$$\frac{d}{dx} \left(\frac{f}{g} \right) = \frac{g \frac{d}{dx} f - f \frac{d}{dx} g}{g^2} \quad (16)$$

Example

Find $h'(x)$ if $h(x) = \tan x$

Recall that $h(x) = \frac{\sin x}{\cos x}$. Hence applying the quotient rule we have

$$\frac{d}{dx} h(x) = \frac{g \frac{d}{dx} f - f \frac{d}{dx} g}{g^2} = \frac{\cos x (\cos x) - \sin x (-\sin x)}{\cos^2 x} = \left(\frac{1}{\cos x} \right)^2 = \sec^2 x$$



Example

Find $h'(x)$ if $h(x) = \frac{x^3 + x + 2}{x^2 + 4}$

Notice that $h(x)$ is the quotient of the functions $f(x) = x^3 + x + 2$ and $g(x) = x^2 + 4$. Hence applying the quotient rule we have

$$\begin{aligned} \frac{d}{dx} h(x) &= \frac{g \frac{d}{dx} f - f \frac{d}{dx} g}{g^2} \\ &= \frac{(x^2 + 4)(3x^2 + 1) - (x^3 + x + 2)(2x)}{(x^2 + 4)^2} \end{aligned}$$



Differentiation of composite functions

- ① Given two function $f(x)$ and $u(x)$, then their composite denoted $h(x)$ is defined by

$$h(x) = f[u(x)] \quad (17)$$

- ② The derivative of $h(x)$ is

$$\frac{d}{dx} h(x) = \frac{df}{du} \cdot \frac{du}{dx} \quad (18)$$

- ③ This is also known as the **chain rule**.

Note

$h(x)$ is not the product of functions. The functions f and u are not multiplying but rather u is "placed" in f as an argument.



Example

Find $h'(x)$ if $h(x) = e^{\cos x}$

- 1 Notice that $h(x)$ is the composite of the functions $f(x) = e^x$ and $u(x) = \cos x$ that is $f(u)$ implies replacing the x (argument) in $f(x)$ by $u(x)$.
- 2 Thus $\frac{df}{du} = e^u$ and $\frac{du}{dx} = -\sin x$.
- 3 Hence applying the composite rule we have

$$\begin{aligned}
 \frac{d}{dx} h(x) &= \frac{df}{du} \cdot \frac{du}{dx} \\
 &= e^u \cdot (-\sin x) \\
 &= -e^{\cos x} \sin x
 \end{aligned}$$



Example

Find $h'(x)$ if $h(x) = \sqrt{x^2 + 5}$

- 1 Notice that $h(x)$ is the composite of the functions $f(u) = \sqrt{u}$ and $u(x) = x^2 + 5$ that is $f(u)$ implies replacing the x (argument) in $f(x)$ by $u(x)$.
- 2 Thus $\frac{df}{du} = \frac{1}{2\sqrt{u}}$ and $\frac{du}{dx} = 2x$.
- 3 Hence applying the composite rule we have

$$\begin{aligned} \frac{d}{dx} h(x) &= \frac{df}{du} \cdot \frac{du}{dx} \\ &= \frac{1}{2\sqrt{u}} \cdot (2x) = \frac{x}{\sqrt{x^2 + 5}} \end{aligned}$$



Example

Find $h'(x)$ if $h(x) = \cos^2(x^3 + 4)$

- 1 Notice that $h(x)$ is the composite function $f(u(z(x)))$ where $f(u) = u^2$, $u(z) = \cos z$ and $z(x) = x^3 + 4$.
- 2 Thus $\frac{df}{du} = 2u$ and $\frac{du}{dz} = -\sin z$ and $\frac{dz}{dx} = 3x$.
- 3 Hence applying the composite rule we have

$$\begin{aligned}
 \frac{d}{dx}h(x) &= \frac{df}{du} \cdot \frac{du}{dz} \cdot \frac{dz}{dx} \\
 &= (2u) \cdot (-\sin z) \cdot (3x) \\
 &= (2\cos(x^3 + 4))(-\sin(x^3 + 4))(3x) \\
 &= -6x \cos(x^3 + 4) \sin(x^3 + 4)
 \end{aligned}$$



Derivatives of inverse functions

Suppose $y = f^{-1}(x)$ then $x = f(y)$. Thus

$$\frac{dy}{dx} = \frac{1}{dx/dy} = \frac{1}{f'(y)} \quad (19)$$

Example

Find the derivatives of

- 1 $y = \ln x$
- 2 $y = \sin^{-1} x$
- 3 $y = \cos^{-1} x$



1

$$y = \ln x \implies x = e^y \quad (20)$$

$$\text{Thus } \frac{dx}{dy} = e^y \quad (21)$$

$$\text{and so } \frac{dy}{dx} = \frac{1}{e^y} = \frac{1}{x} \quad (22)$$

2

$$y = \sin^{-1} x \implies x = \sin y \quad (23)$$

$$\text{Thus } \frac{dx}{dy} = \cos y \quad (24)$$

$$\text{so } \frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1 - \sin^2 y}} = \frac{1}{\sqrt{1 - x^2}} \quad (25)$$

3

$$\frac{d(\cos^{-1} x)}{dx} = \frac{1}{dx/dy} = \frac{1}{-\sin y} = -\frac{1}{\sqrt{1 - \cos^2 y}} = -\frac{1}{\sqrt{1 - x^2}}$$



Implicit Differentiation

Definition (Implicit function)

These are functions which cannot be easily expressed in the form of $y = f(x)$.

Example

The function $y = \frac{1}{2}x^2$ is implicitly defined by the equation $x^2 - 2y = 0$ and the derivative with respect to x is

$$\frac{d}{dx} [x^2 - 2y] = \frac{d}{dx}(0) \quad (26)$$

$$2x - 2y' = 0 \quad (27)$$

$$y' = x \quad (28)$$

Example

The equation $y = \frac{2x+3}{x}$ is implicitly defined as $xy - 2x = 3$ and its differentiation gives

$$\frac{d}{dx} [xy - 2x] = \frac{d}{dx} (3) \quad (29)$$

$$(y + xy') - 2 = 0 \quad (30)$$

$$y' = \frac{2-y}{x} \quad (31)$$

$$y' = \frac{2 - \frac{2x+3}{x}}{x} \quad (32)$$

$$y' = -\frac{3}{x^2} \quad (33)$$



Example

The equation $xy + e^y = 1$ **implicitly** defines the function y . There is no explicit formula to define y . But the derivative with respect to x can be still obtained as follows

$$\frac{d}{dx} [xy + e^y] = \frac{d}{dx} (1) \quad (34)$$

$$(y + xy') + y'e^y = 0 \quad (35)$$

$$(x + e^y)y' = -y \quad (36)$$

$$\frac{dy}{dx} = -\frac{y}{x + e^y} \quad (37)$$

$$= -\frac{y}{x + 1 - xy} \quad (38)$$



Higher-Order derivatives

Notice that the derivatives of functions are also functions. Hence we can find the derivatives of the derivatives of functions. These are precisely the higher derivatives of the given function.

Second order derivatives can be denoted by any of the following

$$1) f''(x), \quad 2) \frac{d^2 f(x)}{dx^2}, \quad 3) D_x^2 f(x), \quad 4) D^2 f(x)$$

Third order derivatives can be denoted by any of the following

$$1) f'''(x), \quad 2) \frac{d^3 f(x)}{dx^3}, \quad 3) D_x^3 f(x), \quad 4) D^3 f(x)$$

n_{th} order derivatives can be denoted by any of the following

$$1) f^n(x), \quad 2) \frac{d^n f(x)}{dx^n}, \quad 3) D_x^n f(x), \quad 4) D^n f(x)$$

Example

$$y = t^3 - 6t^2 + 9t \quad (39)$$

$$\frac{dy}{dx} = 3t^2 - 12t + 9 \quad (40)$$

$$\frac{d^2y}{dx^2} = 6t - 12 \quad (41)$$

$$\frac{d^3y}{dx^3} = 6 \quad (42)$$

$$\frac{d^n y}{dx^n} = 0; \quad n \geq 4 \quad (43)$$



Example

$$y = \sin x \quad (44)$$

$$\frac{dy}{dx} = \cos x \quad (45)$$

$$\frac{d^2 y}{dx^2} = -\sin x \quad (46)$$

$$\frac{d^3 y}{dx^3} = -\cos x \quad (47)$$

$$\frac{d^4 y}{dx^4} = \sin x \quad (48)$$

$$\frac{d^5 y}{dx^5} = \cos x \quad (49)$$

You notice that the higher derivatives does not vanish, but follows some pattern, that is

$$\textcircled{1} \quad f^n(x) = \sin x; \quad n = 4k \text{ for some } k$$

$$\textcircled{2} \quad f^n(x) = \cos x; \quad n = 4k + 1 \text{ for some } k$$

$$\textcircled{3} \quad f^n(x) = -\sin x; \quad n = 4k + 2 \text{ for some } k$$

$$\textcircled{4} \quad f^n(x) = -\cos x; \quad n = 4k + 3 \text{ for some } k$$

Now let look at a theorem that aids in computing the product of higher-order derivatives.



Suppose $y(x)$ is a product of the functions f and g that is $y = fg$. Then

$$Dy = gDf + fDg \quad (50)$$

$$D^2y = D((gDf + fDg)) = gD^2f + 2DfDg + fD^2g \quad (51)$$

$$D^3y = D(gD^2f + 2DfDg + fD^2g) = gD^3f + 3D^2fDg + 3DfD^2g + fD^3g \quad (52)$$

$$\vdots = \vdots$$

$$D^n y = gD^n f + \binom{n}{1} DgD^{n-1}f + \cdots + fD^n g \quad (53)$$

Theorem (Leibnitz theorem)

Let $y = fg$ where f and g are functions of x , then the n th derivative of y is given by

$$D^n y = \sum_{r=0}^n \binom{n}{r} D^r g D^{n-r} f$$

Example

Find the third derivative of $y = x^2 e^x$

① Let $g(x) = x^2 \implies g'(x) = 2x, \quad g''(x) = 2, \quad g'''(x) = 0$

② If $f(x) = e^x \implies f'(x) = f''(x) = f'''(x) = e^x$

$$D^3 y = \sum_{r=0}^3 \binom{3}{r} D^r g D^{3-r} f \quad (54)$$

$$= \binom{3}{0} D^0 g D^3 f + \binom{3}{1} D^1 g D^2 f + \binom{3}{2} D^2 g D^1 f + \binom{3}{3} D^3 g D^0 f \quad (55)$$

$$= x^2 e^x + 3(2x)(e^x) + 3(2)(e^x) + 0(e^x) \quad (56)$$

$$= e^x (x^2 + 6x + 6) \quad (57)$$



Exercise

1 Find the first derivatives of 1) $f(x) = \frac{x^2 + x - 2}{x^3 + 6}$ 2) $f(x) = \frac{e^x}{1 + x^3}$

3) $f(x) = \left(\frac{1}{x^2 + 1} \right) (\tan x)$

2 Find the implicit derivative of the function y or θ with respect to x or r :

1) $y^2 - x^2 = \sin(xy)$

2) $x^3 + y^3 - 9xy = 0$

3) $x^2y + y^2x = 6$

4) $x + \tan(xy) = 0$

5) $x \cos(2x + 3y) = y \sin x$

6) $\cos r + \cot \theta = r$.

3 Find the general formula for the n th derivatives of the following.

1 $y = \ln x$

2 $y = e^{ax}$

3 $y = \sin x$



END OF LECTURE
THANK YOU

